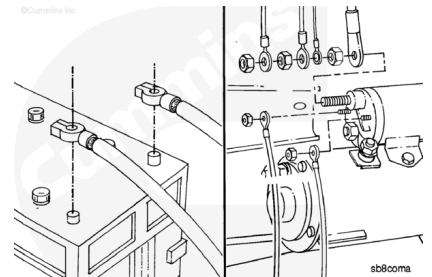


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

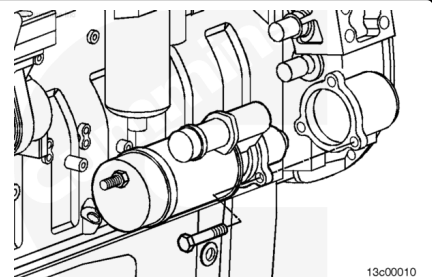


Remove the electrical connections from the batteries.

Remove the electrical connections from the starting motor.

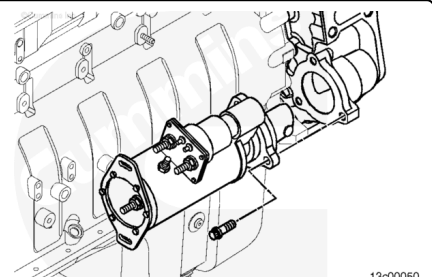
All Except 50MT With Aluminum Flywheel Housing

Remove the three capscrews, starting motor, and starting motor spacer (if used).



50MT With Aluminum Flywheel Housing

Remove the three mounting capscrews, starting motor, and starting motor spacer (if used), as well as the necessary starting motor brackets.

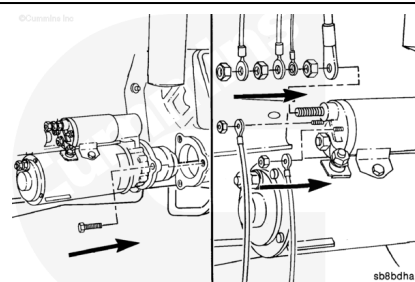


Install

All Except 50MT With Aluminum Flywheel Housing

⚠ CAUTION ⚠

Make sure to use the same thickness of starting motor spacer (if used) as the one removed to install the starting motor to reduce the possibility of engine or starting motor damage.



Install the starting motor, spacer (if applicable) and the three mounting capscrews.

Tighten the mounting capscrews.

All except 39 MT starters:

5/8-11 x 1.75 in capscrew	140 n.m	[103 ft-lb]
5/8-11 x 2.00 in capscrew	201 n.m	[148 ft-lb]

39 MT Starters (Aluminum Flywheel Housing):

	129 n.m	[95 ft-lb]
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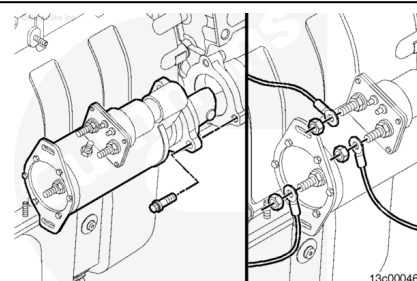
39 MT Starters (Cast Iron Flywheel Housing):

	163 n.m	[120 ft-lb]
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50MT With Aluminum Flywheel Housing

⚠ CAUTION ⚠

Make sure to use the same thickness of starting motor spacer (if used) as the one removed to install the starting motor to reduce the possibility of engine or starting motor damage.



Install the starting motor, spacer (if applicable), and the three mounting capscrews.

Tighten the mounting capscrews.

5/8-11 x 1.75 in capscrew	140 n.m	[103 ft-lb]
5/8-11 x 2.00 in capscrew	201 n.m	[148 ft-lb]

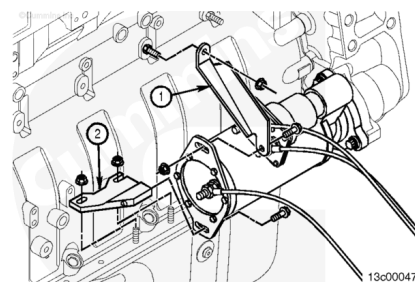
Assemble the upper bracket (1) to the block mounting stud and to the starter motor end bracket. Tighten capscrews and nuts finger tight.

Assemble the lower bracket (2) to the oil pan rail mounting stud and to the starter motor end bracket.

Tighten the capscrews and nuts finger tight.

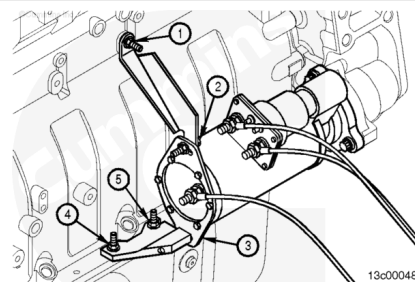
Pre-torque all capscrews and nuts.

Torque Value: 3 n•m [26 in-lb]



Torque all capscrews and nuts in the following sequence:

Location 1	45 n.m	[33 ft-lb]
Location 2	23 n.m	[17 ft-lb]
Location 3	23 n.m	[17 ft-lb]
Location 5	41 n.m	[30 ft-lb]
Location 4	41 n.m	[30 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Install and tighten the electrical connections to the starting motor.

Install the electrical connections to the batteries.

